

Daily Geometry Drill #2 — Answers & Investigations

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Circle Equations · Tangents · Line–Circle Tests · Chord Lengths.

Section A Rapid Recognition

1. Write down the centre of the circle $x^2 + y^2 - 6x - 8y + 24 = 0$.

Answer: "(3, 4)"

Method: Complete the square: $(x - 3)^2 + (y - 4)^2 = 25 - 24 = 1$, so centre (3, 4), radius 1.

Investigate further: In $x^2 + y^2 + 2gx + 2fy + c = 0$ the centre is $(-g, -f)$ and here $2g = -6$, $2f = -8$, so $g = -3$, $f = -4$ — centre (3, 4) ✓. **If the coefficients of x^2, y^2 are not 1, divide through first so they are, then read the centre.**

2. For which value of k does $x^2 + y^2 - 6x + 4y + k = 0$ represent a circle of radius 5?

Answer: -12

Method: Complete the square: $(x - 3)^2 + (y + 2)^2 = 13 - k$. Radius 5 needs $13 - k = 25$, so $k = -12$.

Investigate further: For which k is this a real circle at all, and what does the equation describe at the boundary value?

3. Write down the equation of the circle with centre (2, -1) and radius 5.

Answer: " $x^2 + y^2 - 4x + 2y - 20 = 0$ "

Method: $(x - 2)^2 + (y + 1)^2 = 25$ expands to $x^2 + y^2 - 4x + 2y + 5 = 25$, i.e. $x^2 + y^2 - 4x + 2y - 20 = 0$.

Investigate further: Reading back: $2g = -4 \Rightarrow g = -2$ (centre $x = 2$), $2f = 2 \Rightarrow f = 1$ (centre $y = -1$), $r = \sqrt{g^2 + f^2 - c} = \sqrt{4 + 1 + 20} = 5$ ✓. **Rewrite for centre (2, -1) radius 2: what becomes of the constant?** $x^2 + y^2 - 4x + 2y + 1 = 0$.

4. Write down the radius of the circle $x^2 + y^2 - 4x + 6y - 12 = 0$.

Answer: 5

Method: Complete the square: $(x - 2)^2 + (y + 3)^2 = 4 + 9 + 12 = 25$, so $r = 5$.

Investigate further: Formula: $g = -2, f = 3, r = \sqrt{4 + 9 + 12} = 5$ ✓. **All these squares are perfect: $5^2, 4^2, 3^2$. What radius would an *irrational* answer look like, and would that stop a clean distance test?** Only the appearance changes; the distance test still works.

5. The point $(x, 4)$ lies on the circle $x^2 + y^2 = 25$ with $x > 0$. Find x .

Answer: 3

Method: $x^2 + 4^2 = 25 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$; take +3.

Investigate further: Both (3, 4) and (-3, 4) lie on the circle — the $x > 0$ condition kills one branch.

List all lattice points (x, y) **on** $x^2 + y^2 = 25$. $(0, \pm 5), (\pm 3, \pm 4), (\pm 4, \pm 3), (\pm 5, 0)$.

6. The radius from the origin to the point (3, 4) on $x^2 + y^2 = 25$ has what gradient?

Answer: $\frac{4}{3}$

Method: Gradient = $\frac{4-0}{3-0} = \frac{4}{3}$.

Investigate further: The radius is the line joining centre to the point. **Gradient of the perpendicular radius to** (3, -4)? $-\frac{4}{3}$.

7. The tangent to $x^2 + y^2 = 25$ at the point (3, 4) has what gradient?

Answer: $-\frac{3}{4}$

Method: Tangent $\perp$ radius, and radius gradient is $\frac{4}{3}$, so tangent gradient = $-\frac{1}{4/3} = -\frac{3}{4}$.

Investigate further: The tangent line is $3x + 4y = 25$ (via $xx_1 + yy_1 = r^2$). **Does it pass through** (3, 4)? $9 + 16 = 25$ ✓. **What is its x-intercept?** $x = 25/3$.

8. Write down the equation of the circle with centre (3, -4) that passes through the origin.

Answer: $(x - 3)^2 + (y + 4)^2 = 25$

Method: The radius is the distance from (3, -4) to (0, 0), which is 5.

Investigate further: Expanded, the equation has no constant term. Explain why every circle through the origin has $c = 0$ in the form $x^2 + y^2 + 2gx + 2fy + c = 0$.

9. The centre of $x^2 + y^2 - 6x - 8y + 24 = 0$ is (3, 4). Find the perpendicular distance of this centre from the line $4x + 3y - 9 = 0$.

Answer: 3

Method: Distance = $\frac{|4(3)+3(4)-9|}{\sqrt{4^2+3^2}} = \frac{|12+12-9|}{5} = \frac{15}{5} = 3$.

Investigate further: This is Day 1's perpendicular-distance formula applied to a centre. Since $d = 3 > r = 1$, the line misses the circle: find the line parallel to $4x + 3y - 9 = 0$ that is tangent to the circle on the near side.

10. A chord of the circle $x^2 + y^2 = 25$ is at perpendicular distance 3 from the origin. Find the length of the chord.

Answer: 8

Method: Half-chord = $\sqrt{r^2 - d^2} = \sqrt{25 - 9} = \sqrt{16} = 4$; full length = 8.

Investigate further: The perpendicular from the centre to a chord bisects it — the half-chord formula. **If the distance were 4, the length is 6; if 5, it degenerates to a tangent. Check both.** The chord shrinks to a point exactly when $d = r$.

Section B Manipulation Drills

1. Which of the following is the equation of the circle with centre $(3, -2)$ and radius 4?

- A) $x^2 + y^2 - 6x + 4y - 3 = 0$
 B) $x^2 + y^2 - 6x + 4y + 9 = 0$
 C) $x^2 + y^2 + 6x - 4y - 3 = 0$
 D) $x^2 + y^2 - 6x - 4y - 3 = 0$

Answer: A) $x^2 + y^2 - 6x + 4y - 3 = 0$

Method: $(x - 3)^2 + (y + 2)^2 = 16$ gives $x^2 + y^2 - 6x + 4y + 13 = 16$, i.e. $x^2 + y^2 - 6x + 4y - 3 = 0$.

Investigate further: Only A has both the x -coefficient -6 (centre $x = 3$) and the y -coefficient $+4$ (centre $y = -2$). **B has the right coefficients but the wrong constant $+9$; what radius would that give? $r = \sqrt{9 + 4 - 9} = 2$.**

2. The circle $x^2 + y^2 + 2x - 4y - 11 = 0$ has what radius?

- A) 2
 B) 3
 C) 4
 D) $\sqrt{5}$

Answer: C) 4

Method: $(x + 1)^2 + (y - 2)^2 = 1 + 4 + 11 = 16$, so $r = 4$.

Investigate further: Complete the square exactly: $x^2 + 2x = (x + 1)^2 - 1$ and $y^2 - 4y = (y - 2)^2 - 4$, so $(x + 1)^2 + (y - 2)^2 = 11 + 1 + 4 = 16$ and $r = 4$ ✓. With $g = 1, f = -2, c = -11$ the formula reads $r = \sqrt{g^2 + f^2 - c} = \sqrt{1 + 4 + 11}$. **Option D ($\sqrt{5}$) traps students who drop the constant 11 (leaving $r^2 = 1 + 4 = 5$) or flip the sign of c ; keep the sign fixed through the whole square.**

3. The point $(1, 2)$ lies on the circle $x^2 + y^2 = 5$. The equation of the tangent to the circle at $(1, 2)$ is?

- A) $y = -\frac{1}{2}x + \frac{5}{2}$
 B) $y = 2x$
 C) $y = -2x + 5$
 D) $y = \frac{1}{2}x$

Answer: A) $y = -\frac{1}{2}x + \frac{5}{2}$

Method: Radius gradient $= \frac{2}{1} = 2$; tangent gradient $= -\frac{1}{2}$; through $(1, 2)$: $y - 2 = -\frac{1}{2}(x - 1) \Rightarrow y = -\frac{1}{2}x + \frac{5}{2}$.

Investigate further: Option B is the radius line itself, option C negates the radius gradient without inverting it, and option D inverts without negating. The split-variable form $xx_1 + yy_1 = r^2$ gives $x + 2y = 5$ directly.

4. A circle of form $x^2 + y^2 + 2gx + 2fy + c = 0$ passes through the origin. Deduce c .

- A) $c = 1$
 B) $c = 0$
 C) $c = g + f$
 D) $c = -1$

Answer: B) $c = 0$

Method: At the origin $(0, 0) \in$ circle means $0 + 0 + 0 + 0 + c = 0$ — the constant term is the value of the LHS at the origin, so $c = 0$.

Investigate further: Circle $x^2 + y^2 = 25$ has $c = -25$ and indeed does *not* pass through the origin. The constant records the origin's membership: $c = 0 \Leftrightarrow$ origin on the circle.

5. The perpendicular distance from the centre of $x^2 + y^2 = 25$ to the line $y = 2x + 3$ is?

- A) $\frac{3}{\sqrt{5}}$
 B) $\frac{3}{5}$
 C) 3
 D) 2

Answer: A) $\frac{3}{\sqrt{5}}$

Method: Line $2x - y + 3 = 0$; distance from $(0, 0) = \frac{|3|}{\sqrt{2^2 + (-1)^2}} = \frac{3}{\sqrt{5}}$.

Investigate further: Compare with $r = 5$: $\frac{3}{\sqrt{5}} < 5$, so the line cuts the circle. This is exactly the Day-1 formula put to the new line-circle test.

6. Which of the following is a tangent to the circle $x^2 + y^2 = 4$?

- A) $y = x + 2\sqrt{2}$
 B) $y = x + 4$
 C) $y = 2x + 2$
 D) $y = 2x + 4$

Answer: A) $y = x + 2\sqrt{2}$

Method: Line $x - y + 2\sqrt{2} = 0$ has distance from origin $\frac{2\sqrt{2}}{\sqrt{2}} = 2 = r$ — tangent. The others give $\frac{4}{\sqrt{2}} = 2\sqrt{2}$, $\frac{2}{\sqrt{5}}$, $\frac{4}{\sqrt{5}}$, none equal to 2.

Investigate further: Only option A places the origin exactly r away. **Check B:** $y = x + 4$ has distance $2\sqrt{2} > 2$, so it misses the circle. Shifting a tangent line a little breaks tangency — distance must *equal* r .

7. A chord of the circle $x^2 + y^2 = 13$ has length $4\sqrt{3}$. What is the perpendicular distance of the chord from the centre?

- A) 3
- B) 2
- C) $\sqrt{5}$
- D) 1

Answer: D) 1

Method: Half-chord = $2\sqrt{3}$; $(2\sqrt{3})^2 = 13 - d^2 \Rightarrow 12 = 13 - d^2 \Rightarrow d^2 = 1 \Rightarrow d = 1$.

Investigate further: Fallback: remember (half-chord)² + ($\perp$ dist)² = r^2 . Statement C ($\sqrt{5}$) is the trap of $13 - 8$ instead of $13 - 12$.

8. The line $y = mx + 4$ is a tangent to the circle $x^2 + y^2 = 4$. Find the two possible values of m .

- A) ± 1
- B) $\pm\sqrt{3}$
- C) ± 2
- D) $\pm\sqrt{2}$

Answer: B) $\pm\sqrt{3}$

Method: Line $mx - y + 4 = 0$; tangency requires $\frac{|4|}{\sqrt{m^2+1}} = 2 \Rightarrow \sqrt{m^2+1} = 2 \Rightarrow m^2 = 3 \Rightarrow m = \pm\sqrt{3}$.

Investigate further: Both tangent lines pass through $(0, 4)$, symmetric about the y -axis — the two tangents from an external point. **How long is each tangent segment from $(0, 4)$ to its touch point?** $\sqrt{16 - 4} = \sqrt{12} = 2\sqrt{3}$ — a preview of power of a point (Day 6).

9. The circle with centre $(6, a)$ and radius 6 is tangent to the y -axis and passes through the point $(0, 8)$. Find a .

- A) 1
- B) 3
- C) 5
- D) 8

Answer: D) $a = 8$

Method: Tangent to the y -axis ($x = 0$) since centre's x -distance $6 = r$. Through $(0, 8)$: $(0-6)^2 + (8-a)^2 = 36 \Rightarrow 36 + (8-a)^2 = 36 \Rightarrow a = 8$.

Investigate further: The circle touches the y -axis at $(0, a)$, and it also passes through $(0, 8)$. A line meets a circle it is tangent to only once, so those points must coincide: $a = 8$ with no algebra.

10. Write down the length of the chord cut from the line $y = x + 1$ by the circle $x^2 + y^2 = 25$.

Answer: $7\sqrt{2}$

Method: Line $x - y + 1 = 0$; distance from the origin = $\frac{|1|}{\sqrt{2}} = \frac{1}{\sqrt{2}}$. Half-chord = $\sqrt{25 - \frac{1}{2}} = \sqrt{\frac{49}{2}} = \frac{7}{\sqrt{2}}$; full chord = $\frac{14}{\sqrt{2}} = 7\sqrt{2}$.

Investigate further: Note the line does not pass through the centre, so this is a genuine off-centre chord. If it were $y = x$ the chord would be a full diameter 10.

Section C Substitution & Structure

(≤90 s each)

Harder: hidden tangency, chord bisection, and the distance-vs-discriminant decision. No brute substitution.

1. A tangent to the circle $x^2 + y^2 = 144$ passes through $(20, 0)$ and meets the positive y -axis at T . What is the y -coordinate of T ? (after TMUA 2017 Paper 1 Q6)

- A) 12
B) 15
C) $\frac{49}{3}$
D) 20

Answer: B) 15

Method: Radius to touch point is $\perp$ tangent. Let $T = (0, t)$, line through $(20, 0)$ and $(0, t)$ has equation $x/20 + y/t = 1$, i.e. $tx + 20y - 20t = 0$. Distance from origin = $r = 12$: $|-20t|/\sqrt{t^2 + 400} = 12 \Rightarrow 400t^2 = 144(t^2 + 400) \Rightarrow 256t^2 = 57600 \Rightarrow t^2 = 225 \Rightarrow t = 15$ (positive). Check $t = 15$ gives line $3x + 4y = 60$, distance $60/5 = 12 \checkmark$.

Investigate further: The tangent from $(20, 0)$ touches at a point X with $OX = 12$ and $OX \perp XT$, so $OX \perp XT$ is a 12-16-20 right triangle (a scaled 3-4-5). Use similar triangles to get $OT = 15$ without any algebra.

2. The point $P(8, 6)$ is outside the circle $x^2 + y^2 = 36$. Tangents from P meet the circle at A and B . What is the length AB ? (adapted from TMUA 2018 Paper 1 Q7)

- A) $\frac{12\sqrt{13}}{13}$
B) $\frac{24\sqrt{13}}{13}$
C) $\frac{48}{5}$
D) $\frac{72}{5}$

Answer: C) $\frac{48}{5}$

Method: $OP = \sqrt{64 + 36} = 10$, $r = 6$, right triangle OAP has $AP = \sqrt{100 - 36} = 8$, distance from O to chord AB is $d = r^2/OP = 36/10 = 18/5$. Half-chord = $\sqrt{r^2 - d^2} = \sqrt{36 - 324/25} = 24/5$, so $AB = 48/5$.

Investigate further: The chord of contact from P is the line $8x + 6y = 36$ (the polar of P). Check that its distance from O is $\frac{18}{5}$, and show that in general $AB = \frac{2r\sqrt{OP^2 - r^2}}{OP}$.

3. The point Q moves on the circle $x^2 + y^2 - 2x + 4y - 20 = 0$, and P is the point $(13, 3)$. What is the least possible distance PQ ?

- A) 13
B) 18
C) 12

D) 8

Answer: D) 8

Method: Centre $(1, -2)$, radius 5. P is $\sqrt{12^2 + 5^2} = 13$ from the centre, so the nearest point of the circle is $13 - 5 = 8$ away, on the line from P to the centre. Option A is the distance to the centre; option B is the *greatest* distance; option C is the tangent length $\sqrt{13^2 - 5^2}$.

Investigate further: The tangent length (12) lies between the least and greatest distances (8 and 18). Show that it is always their geometric mean.

4. A circle has equation $x^2 + y^2 - 18x - 22y + 178 = 0$. A regular hexagon $ABCDEF$ is inscribed so that all its vertices lie on the circle. What is the area of the hexagon? (*adapted from TMUA 2017 Paper 1 Q9*)

A) $18\sqrt{3}$ B) $36\sqrt{3}$ C) $\frac{75\sqrt{3}}{2}$

D) 75

Answer: B) $36\sqrt{3}$

Method: Complete the square: $r^2 = 9^2 + 11^2 - 178 = 24$. A regular hexagon is six equilateral triangles of side r , so its area is $\frac{3\sqrt{3}}{2}r^2 = \frac{3\sqrt{3}}{2} \cdot 24 = 36\sqrt{3}$. Option C assumes $r = 5$ (a tempting guess from 9, 11); option A uses three triangles instead of six.

Investigate further: The centre $(9, 11)$ never matters: translating a shape does not change its area. What is the area of the square inscribed in the same circle?

5. For which values of p does $x^2 - 2px + y^2 - 6y - p^2 + 8p + 9 = 0$ represent a (real) circle? (*after TMUA 2022 Paper 1 Q2*)

A) $p < 0$ or $p > 4$ B) $-1 < p < 9$ C) $0 < p < 4$ D) $p < -1$ or $p > 9$ **Answer:** A) $p < 0$ or $p > 4$

Method: Complete the squares: $(x - p)^2 + (y - 3)^2 = 2p^2 - 8p = 2p(p - 4)$. A real circle needs $2p(p - 4) > 0$, so $p < 0$ or $p > 4$. At $p = 0$ and $p = 4$ it is a single point. Option C is the interval where it is *not* a circle.

Investigate further: As p varies, where do the centres lie, and what is the smallest radius the circle can have when $p \geq 5$?

6. The circle $x^2 + y^2 - 4x + 2y - 11 = 0$ and the line $x - y + 1 = 0$: determine the number of intersection points without solving the system. (*after MAT 2019 Q6*)

A) 0 (line misses)

B) 1 (tangent)

- C) 2 (secant)
 D) It cannot be decided without solving the system

Answer: C) 2 (secant)

Method: Centre $(2, -1)$, $r = 4$, distance to $x - y + 1 = 0$ is $|2 + 1 + 1|/\sqrt{2} = 4/\sqrt{2} = 2\sqrt{2} < r$, so 2 points.

Investigate further: The line $x - y + 1 = 0$ passes through the centre only if $2 + 1 + 1 = 0$, which fails. Even a line through the centre meets the circle in exactly two points: find them for $x - y - 3 = 0$.

7. The circle $x^2 + y^2 - 8x - 4y - 21 = 0$ is cut by the y -axis. What is the length of the chord intercepted on the y -axis?

- A) $2\sqrt{21}$
 B) $2\sqrt{41}$
 C) 8
 D) 10

Answer: D) 10

Method: Centre $(4, 2)$, $r^2 = 16 + 4 + 21 = 41$. The y -axis is 4 from the centre, so the half-chord is $\sqrt{41 - 16} = 5$ and the chord is 10. Check: $x = 0$ gives $y^2 - 4y - 21 = 0$, so $y = 7$ or -3 . Option A uses $\sqrt{c}$; option B is the diameter.

Investigate further: The product of the two y -intercepts is -21 , the constant term. Explain why, and use it to find the intercept product for any circle $x^2 + y^2 + 2gx + 2fy + c = 0$.

8. The line $y = kx + 5$ is tangent to the circle $x^2 + y^2 - 6x - 8y + 24 = 0$. How many distinct values of k satisfy this? (after TMUA 2020 Paper 1 Q7)

- A) 0
 B) 1
 C) 2
 D) Infinitely many

Answer: C) 2

Method: Centre $(3, 4)$, $r = 1$. Every line $y = kx + 5$ passes through $(0, 5)$, which is $\sqrt{10} > 1$ from the centre, so it is outside the circle and exactly two tangents pass through it. (Solving $\frac{|3k+1|}{\sqrt{k^2+1}} = 1$ gives $k = 0$ or $k = -\frac{3}{4}$.) The vertical tangent is not of the form $y = kx + 5$, so it cannot add a third value.

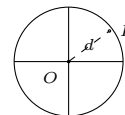
Investigate further: Change the 5 to c . For which values of c is there exactly one value of k ?

Section D Challenge

(TMUA / SMC / BMO1 difficulty)

Hardest 5: distance invariants, hidden right angles, and a construction you must spot.

1. The circles $C_1 : (x+2)^2 + (y-3)^2 = 18$ and $C_2 : (x-7)^2 + (y+6)^2 = 2$ have centres O_1, O_2 . What is the *shortest* distance between a point on C_1 and a point on C_2 ? (adapted from TMUA 2018 Paper 1 Q7 / 2022 Paper 1 Q3)



- A) $5\sqrt{2} - 4$
 B) $5\sqrt{2} - 5$
 C) $5\sqrt{2}$
 D) $5\sqrt{2} + 5$

Answer: C) $5\sqrt{2}$

Method: $O_1 = (-2, 3)$, $O_2 = (7, -6)$, $O_1O_2 = \sqrt{9^2 + 9^2} = 9\sqrt{2}$, $r_1 = 3\sqrt{2}$, $r_2 = \sqrt{2}$, sum $4\sqrt{2}$, shortest $= 9\sqrt{2} - 4\sqrt{2} = 5\sqrt{2}$. Externally separated ($d > r_1 + r_2$).

Investigate further: Shortest = centre distance minus sum of radii (external). If one circle contained the other, it would be $|r_1 - r_2| = d$. For these numbers, $5\sqrt{2} \approx 7.07$.

2. The segment joining $(3, 3)$ and $(7, 5)$ is a diameter of circle C . C is translated 3 left, reflected in the x -axis, then enlarged scale factor 4 about its centre. What is the equation of the final circle? (after TMUA Specimen Paper 1 Q9)

- A) $(x - 2)^2 + (y - 4)^2 = 80$
 B) $(x - 2)^2 + (y + 4)^2 = 80$
 C) $(x - 2)^2 + (y - 4)^2 = 320$
 D) $(x - 2)^2 + (y + 4)^2 = 320$

Answer: B) $(x - 2)^2 + (y + 4)^2 = 80$

Method: Original centre $(5, 4)$, $r^2 = (2)^2 + 1^2 = 5$. Translate 3 left: $(2, 4)$, reflect in x -axis: $(2, -4)$, enlarge $4\times$ about centre: $r^2 = 5 \cdot 16 = 80$, so $(x - 2)^2 + (y + 4)^2 = 80$.

Investigate further: Enlargement about the centre leaves the centre fixed and multiplies the radius by 4, so r^2 by 16. The 320 options scale r^2 by 64 (radius by 8). Where would the centre go if the enlargement were about the origin instead?

3. A circle has centre O and radius 6. Points P and Q lie on it with $\angle POQ \geq 90^\circ$, and triangle POQ has area $9\sqrt{3}$. The point R moves on the major arc PQ . What is the greatest possible area of triangle PRQ ? (adapted from TMUA 2022 Paper 1 Q14)

- A) $18 + 9\sqrt{3}$
 B) $27\sqrt{3}$
 C) $27 + 9\sqrt{3}$
 D) $36 + 9\sqrt{3}$

Answer: B) $27\sqrt{3}$

Method: $[POQ] = \frac{1}{2} \cdot 36 \sin \theta = 9\sqrt{3}$ gives $\sin \theta = \frac{\sqrt{3}}{2}$, and $\theta \geq 90^\circ$ forces $\theta = 120^\circ$. Then $PQ = 6\sqrt{3}$ and O is 3 from PQ . R is farthest from PQ at the far end of the diameter through the midpoint, at height $6 + 3 = 9$, so the area is $\frac{1}{2} \cdot 6\sqrt{3} \cdot 9 = 27\sqrt{3}$. Option A is the answer if you take $\theta = 60^\circ$.

Investigate further: The condition $\angle POQ \geq 90^\circ$ is doing real work: $\sin 60^\circ = \sin 120^\circ$. Check that $\theta = 60^\circ$ gives the same $[POQ]$ but a greatest area of $18 + 9\sqrt{3}$.

4. Two tangents are drawn from the origin to the circle $(x - 5)^2 + (y - 5)^2 = 5$. What is the tangent of the acute angle between them?

- A) $\frac{3}{4}$
 B) $\frac{4}{3}$
 C) $\frac{5}{2}$
 D) 1

Answer: A) $\frac{3}{4}$

Method: The origin is $\sqrt{50}$ from the centre and $r = \sqrt{5}$, so half the angle has $\sin \frac{\theta}{2} = \frac{1}{\sqrt{10}}$ and $\tan \frac{\theta}{2} = \frac{1}{3}$. Then $\tan \theta = \frac{2/3}{1 - 1/9} = \frac{3}{4}$. Check: the tangents $y = mx$ need $\frac{|5m-5|}{\sqrt{m^2+1}} = \sqrt{5}$, so $2m^2 - 5m + 2 = 0$ and $m = 2$ or $\frac{1}{2}$, and $\frac{2-1/2}{1+1} = \frac{3}{4}$. Option C is the sum of the gradients, option D their product, option B the reciprocal.

Investigate further: The two gradients multiply to 1. What does that say about how the two tangents sit relative to the line $y = x$, and why does the symmetry of the circle force it?

5. The circle through $(0, 0)$, $(8, 0)$ and $(2, 6)$ meets the y -axis again at Q . How far is Q from the origin?

- A) $2\sqrt{5}$
 B) 6
 C) 2
 D) 4

Answer: D) 4

Method: The centre is on $x = 4$ (perpendicular bisector of the first two points). Equidistance from $(0, 0)$ and $(2, 6)$: $16 + b^2 = 4 + (6 - b)^2$ gives $b = 2$, so the centre is $(4, 2)$ and $r^2 = 20$. On $x = 0$: $(y - 2)^2 = 4$, so $y = 0$ or 4 , and $Q = (0, 4)$. Option A is the radius; option C is the centre's height.

Investigate further: Get OQ without the centre: the power of the origin is zero, so use the equation $x^2 + y^2 + 2gx + 2fy = 0$ and substitute the other two points. Which coefficient gives OQ directly?

Top 5 Patterns — Worksheet #2

- Complete the square before anything else.** Centre $(-g, -f)$, $r^2 = g^2 + f^2 - c$; never trust a radius you have not computed. (see A1, A2, B2, C4, C5)
- Tangency is a distance statement.** A line is tangent exactly when the centre is r from it; never substitute the line into the circle. (see B6, B8, C8, D4)
- Tangent $\perp$ radius.** At a point of the circle, the tangent's gradient is the negative reciprocal of the radius's; on $x^2 + y^2 = r^2$ it is $xx_1 + yy_1 = r^2$. (see A7, B3, C1)

4. **Half-chord** $= \sqrt{r^2 - d^2}$. The perpendicular from the centre bisects every chord. (see A10, B7, B10, C7)
5. **Go through the centre.** Nearest and farthest points of a circle, and the gap between two circles, all lie on lines through centres. (see C3, D1, D3)

Common Traps — Worksheet #2

1. **Guessing the radius.** Nice-looking centres do not mean a nice radius: $x^2 + y^2 - 18x - 22y + 178 = 0$ has $r^2 = 24$. (see C4)
2. **Stopping at the centre.** The distance to the centre is not the distance to the circle; subtract (or add) r . (see C3, D1)
3. **Confusing the gradients with the angle.** The sum or product of the two tangent gradients is not the angle between them. (see D4)
4. **$\sin \theta$ has two angles.** $\sin 60^\circ = \sin 120^\circ$; read every condition in the question before choosing. (see D3)
5. **Dropping the constant.** The constant term decides the radius, whether the circle is real, and whether it passes through the origin. (see A2, B2, B4, C5)

“Tangency is a distance statement in disguise: compare d to r , never solve the intersection.”

Found a mistake or want to contribute a sheet?
 Visit the open-source project at github.com/The-CerealDev/speed-maths